Systemadministration

Introduktion

Hårdvara - Mjukvara

Agenda

Interaktion mellan hårdvara - mjukvara

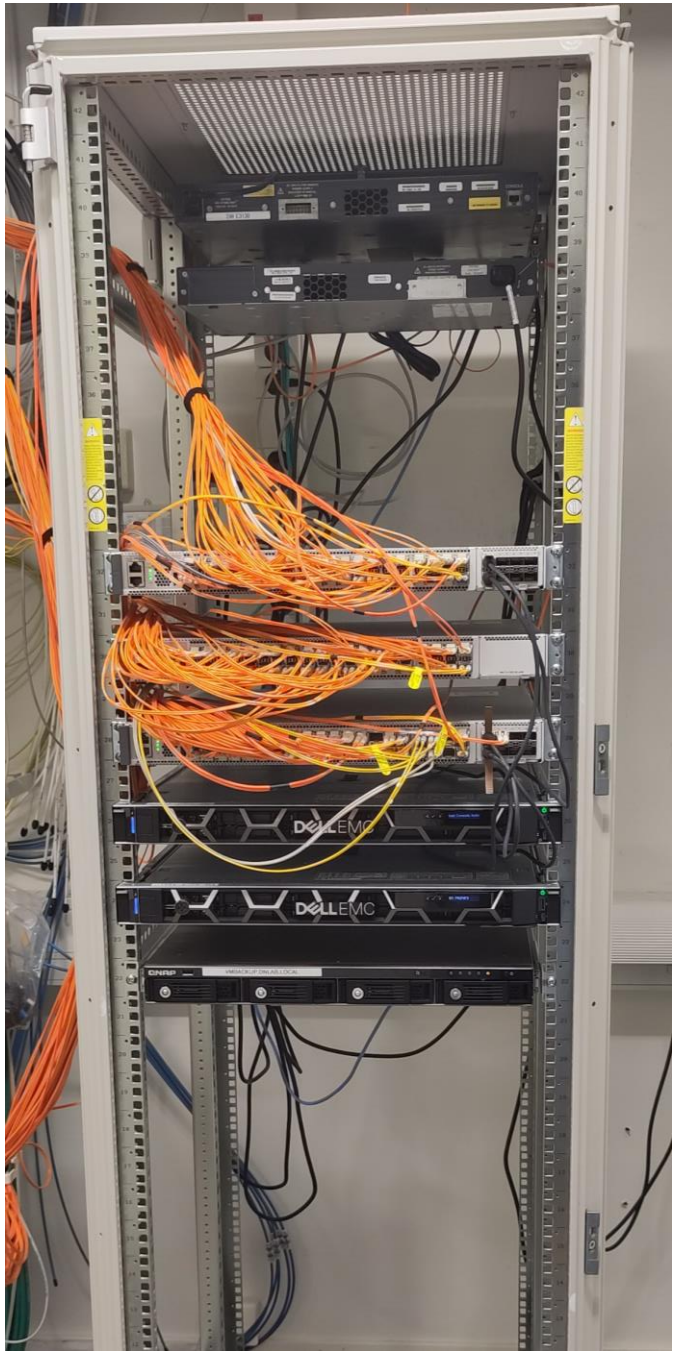
1. Hantering av servrar
2. Virtual Memory
3. Green IT
4. Demo
5. Cybersäkerhet

1. Hantering av servrar

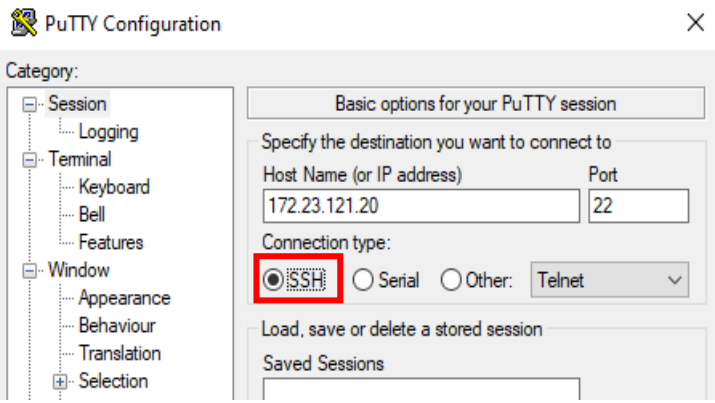
Anslutning

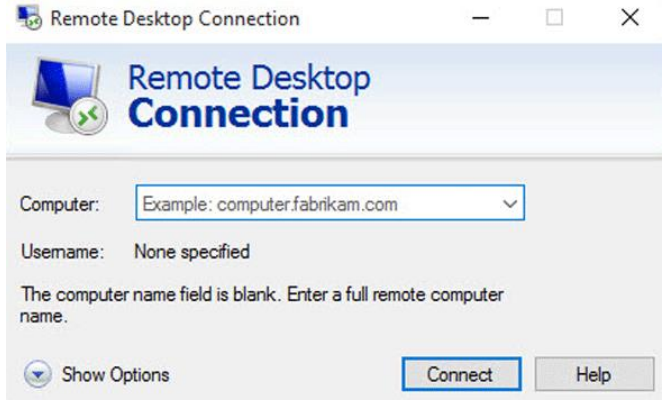
Det finns inga skärmar…

Så hur konfigurerar man en server?



Remote Administration





CLI

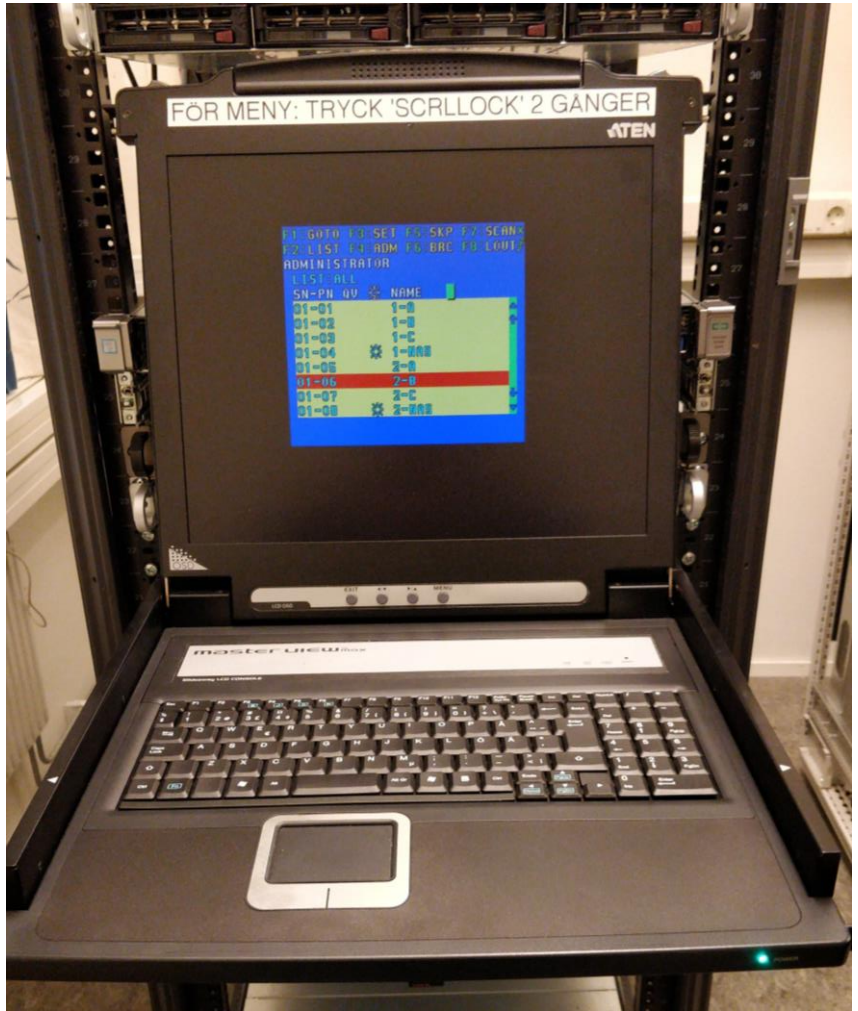
- SSH
- Telnet

GUI

- RDP
- HTTPS
- X-forwarding/Remote X
- SPICE
- VNC

Keyboard Video Mouse

KVM-Console

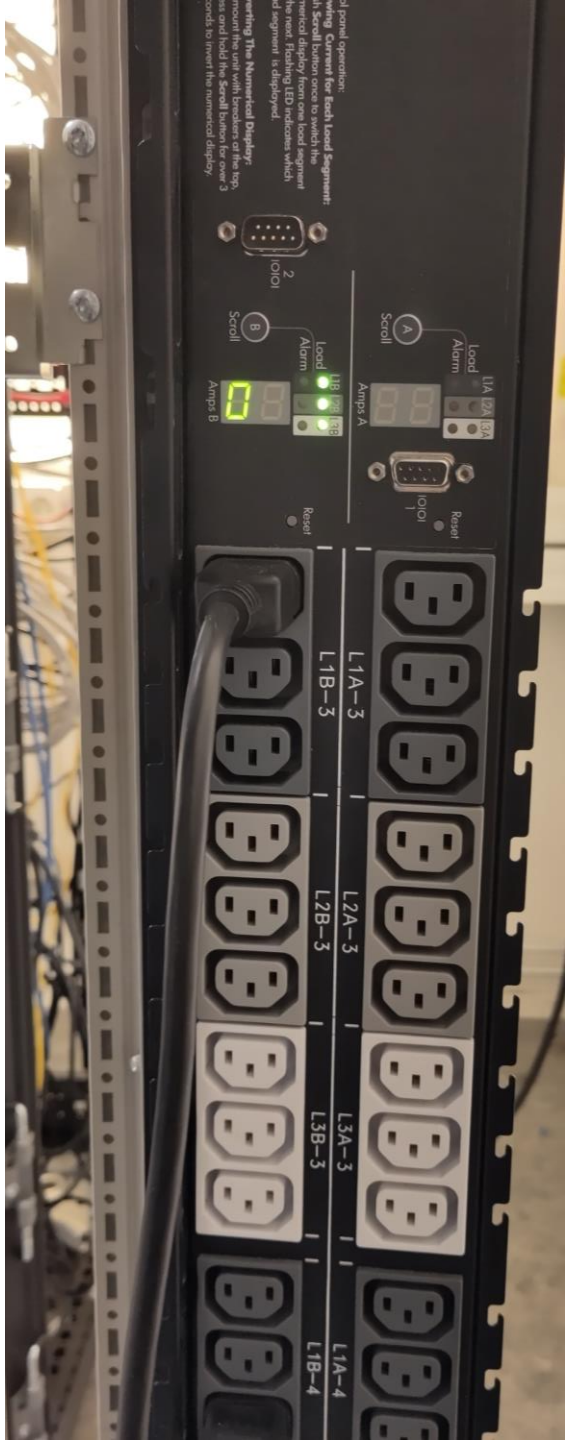


KVM-Switch

Chassis Plans Industrial KVM w/ DVI and USB, by David Lippincott (Rackmount-guy) for Chassis Plans, 2009, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Chassis-Plans-KVM.jpg). CC BY-SA 3.0 Unported.

Power **D**istribution **U**nit

PDU: Förgreningsuttag för servrar

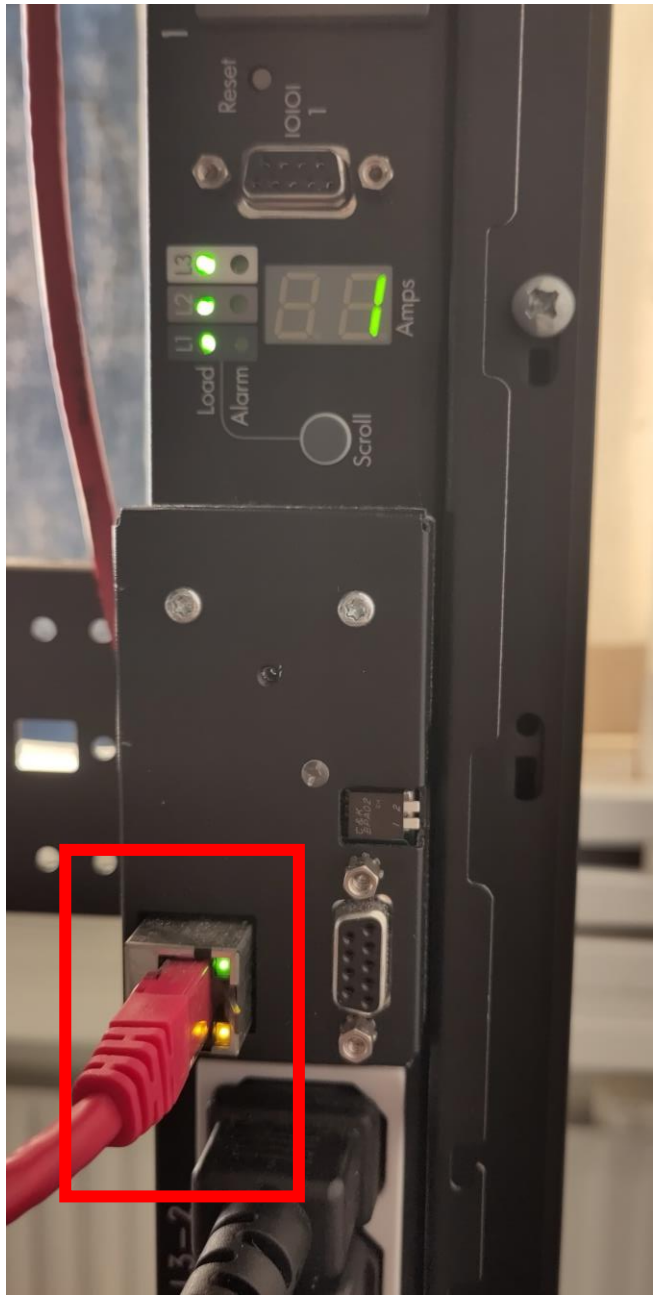


Power **D**istribution **U**nit

PDU: Förgreningsuttag för servrar

Ofta med nätverksanslutning

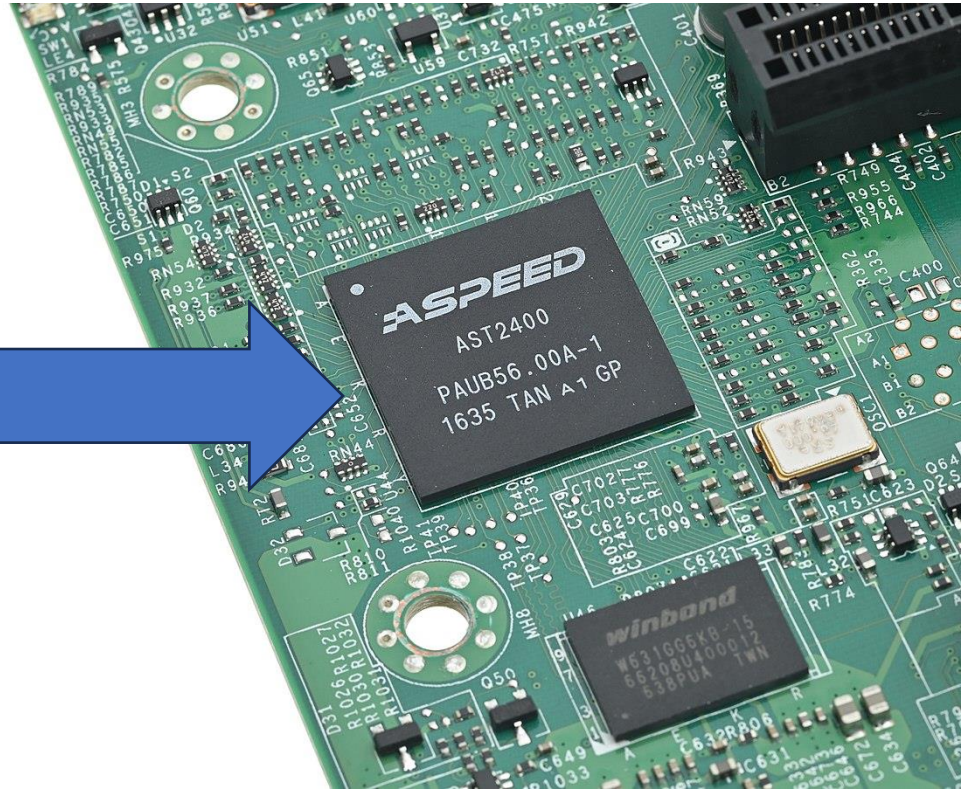
- Fjärrstyrning
- Övervakning



Intelligent Platform Management Interface

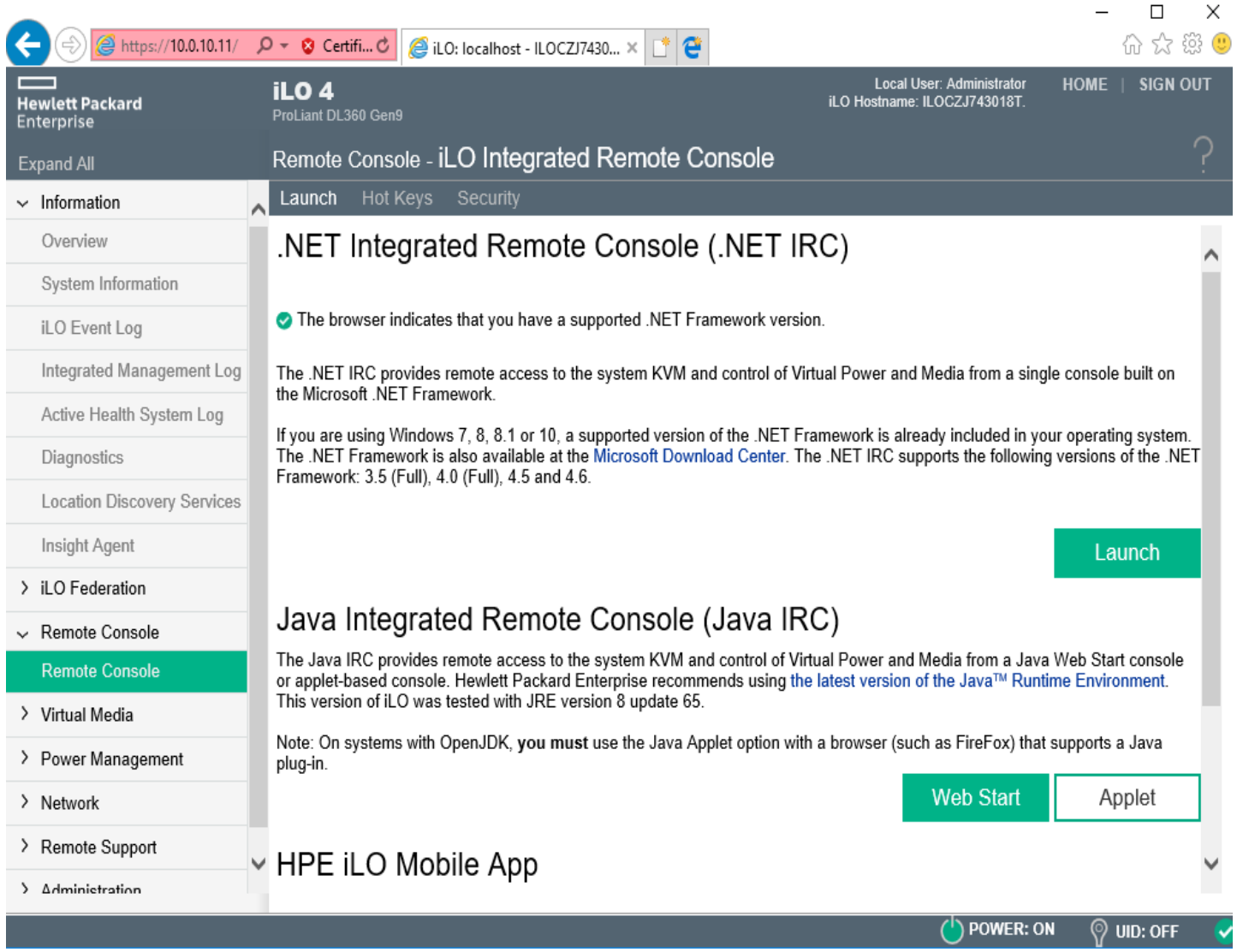
Ett webbgränssnitt för att fjärrstyra en server

Körs på ett BMC-chip



ASPEED AST2400 BMC on a server motherboard, by Phiarc, 2022, Wikimedia Commons (https://en.wikipedia.org/wiki/File:ASPEED_AST2400_BMC_Baseboard_management_controller.jpg). CC BY-SA 4.0 International.

Intelligent Platform Management Interface



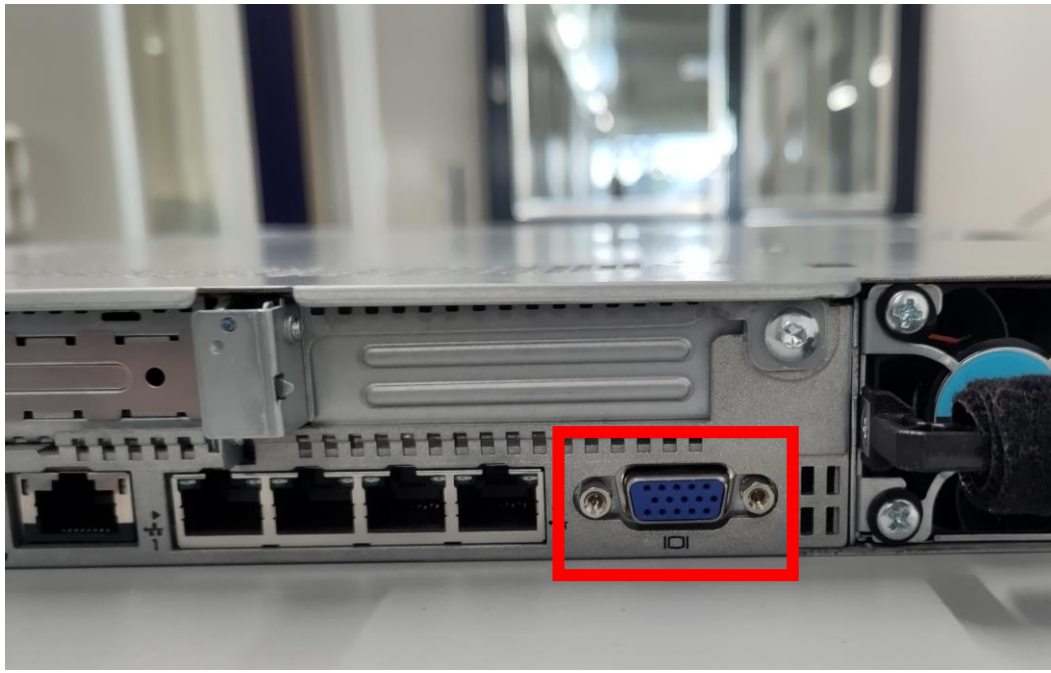
VGA?!

Existerande utrustning är designat för VGA

Enkelt: Analogt & seriellt

→ VGA KVM-switch

→ VGA splitter: BMC & video-output



2. Virtual Memory

Input/Output

Top view of an Intel central processing unit Celeron Kaby Lake type core, model G3930, launched in 2017. LGA 1151 socket, 14 nm process, core frequency 2.90 GHz, Thermal Design Power 51 W, dimensions 37.5mm x 37.5mm. Manufactured in Vietnam. by Erica Gaba – Wikimedia Commons user: Sting, 2022, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=123062267). CC BY-SA 4.0

Beräkningar

MT36JSF1G72PZ-1G9K1HE; Memory Type DDR3 SDRAM; Capacity 8GB; Data Transfer Rate 1866MHz; Pins 240; Bus Type PC-14900; Error Correction Registered ECC; Cycle Time 1.07ns; Cas CL13; Memory Clock 233MHz; Voltage 1.5V; Manufacturer Micron Technology, Inc., by Mister rf, 2023, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=129056892). CC BY-SA 4.0

I/O operations

read/write

Vad händer om RAM-minnet fylls?

För att använda data måste den först läsas in i primärminnet

Processorn kan inte direkt komma åt data på sekundärminnet

Swapping

Data **pages** i RAM måste **swappas** ut mot data på disk

Swap partition

Paging

Data **pages** i RAM måste **swappas** ut mot data på disk

C:\pagefile.sys

Virtual Memory Performance

Top view of an Intel central processing unit Celeron Kaby Lake type core, model G3930, launched in 2017. LGA 1151 socket, 14 nm process, core frequency 2.90 GHz, Thermal Design Power 51 W, dimensions 37.5mm x 37.5mm. Manufactured in Vietnam. by Erica Gaba – Wikimedia Commons user: Sting, 2022, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=123062267). CC BY-SA 4.0

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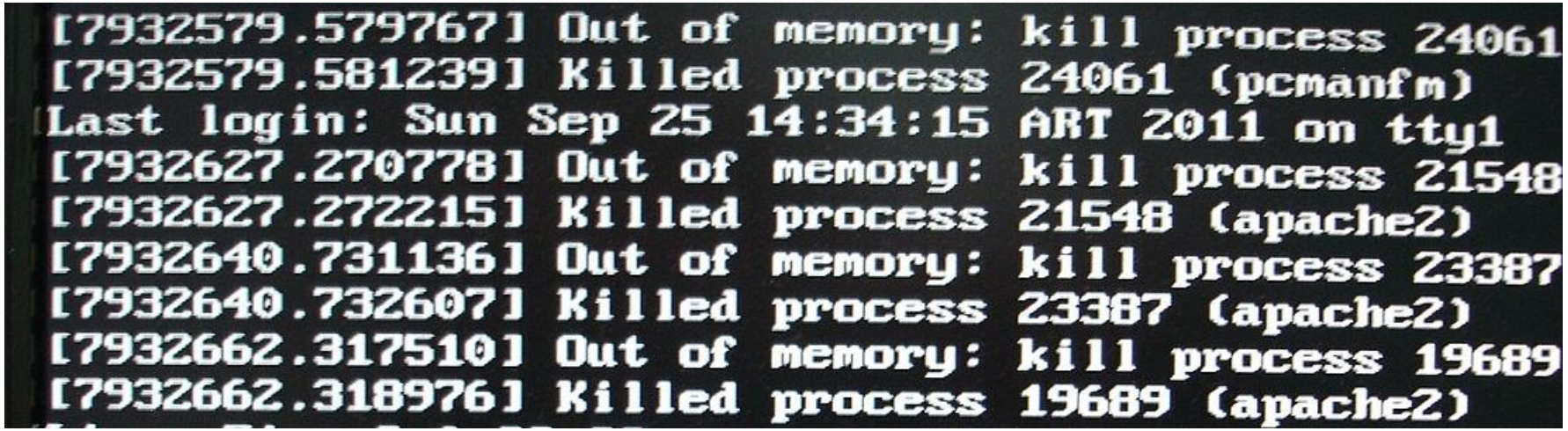
Out Of Memory Killer

Istället för att krascha avslutar **OOM Killer** en/flera processer

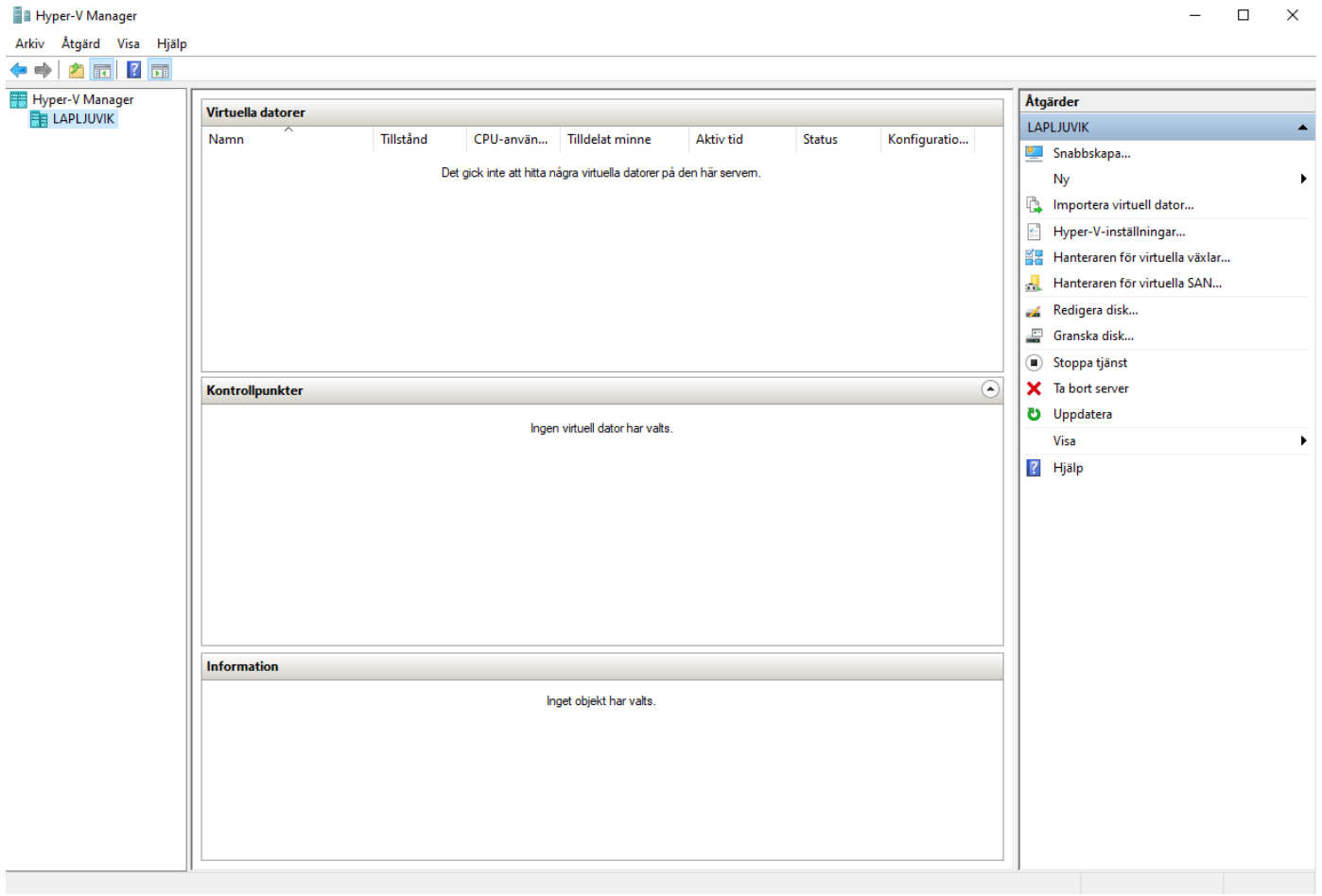


Real Out of Memory scenario in Linux. As you can see, Load was so high that kernell killed two processes before continue printing welcome screen, by Neo139, Wikimedia Commons (https://en.wikipedia.org/wiki/File:Out_of_Memory_in_Linux.jpg). CC BY-SA 3.0 Unported

3. Green IT

Virtualization

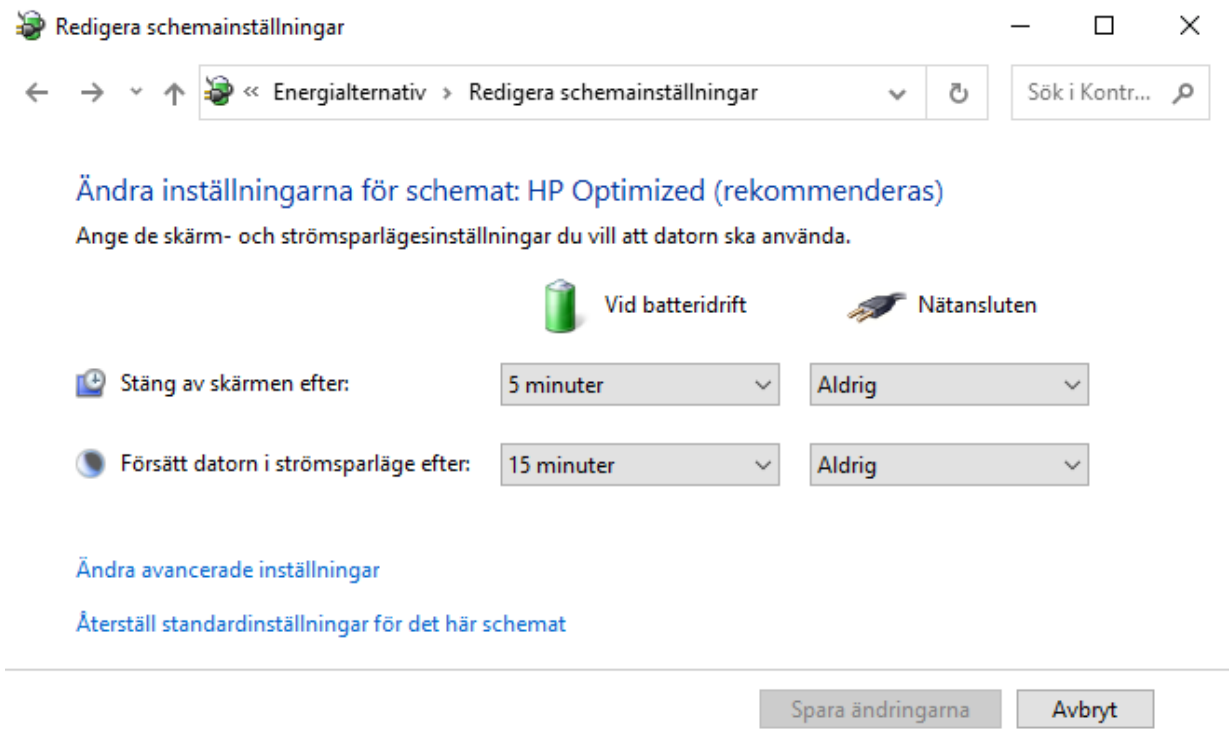
Virtualisering



Windows Power Plan

Styr bland annat:

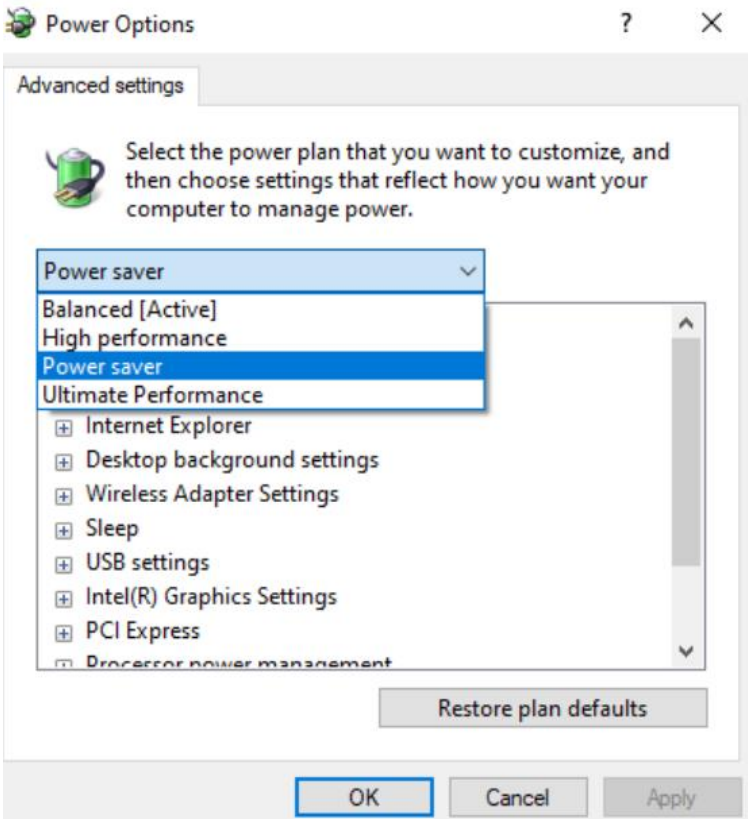
- **Disk parking**
- **CPU clock speed**
- **Sleep**
- **Hibernation**



Windows Power Plan

När borde man använda?

- *Power Saver*
- *High-/ Ultimate Performance*



Sleep

Strömsparläge

Sleep – replacement for the standby power symbol by IEEE 1621, by László Németh, 2014, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Power_sleep_2mm_cropped.svg). Public domain (CC0 1.0 Universal)

Sleep, **standby** eller **Suspend-to-RAM**

- CPU cacher töms till RAM
- Drar minimalt med ström

Hibernate

Viloläge

Hibernate eller **Suspend-to-Disk**

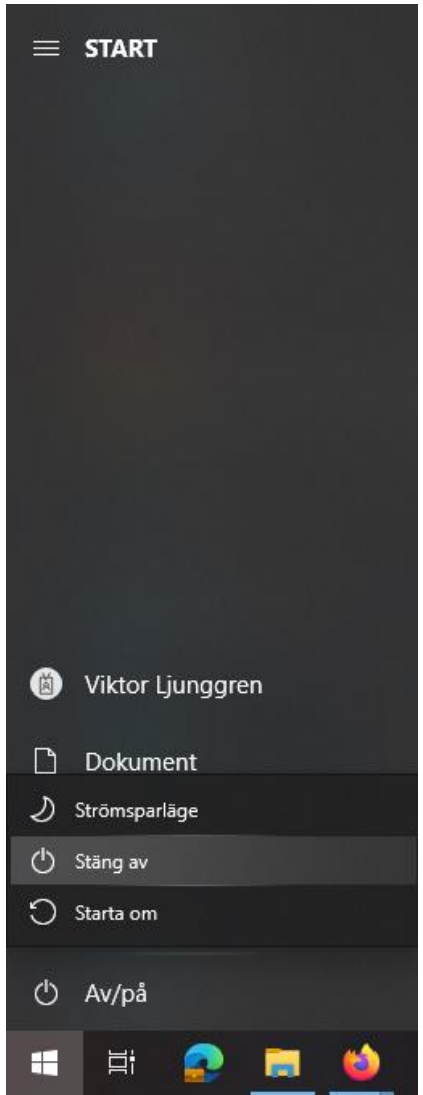
- RAM töms till disk
- Datorn stängs av
- Vid uppstart läses föregående session in från disk

Advanced **C**onfiguration & **P**ower **I**nterface

Strömsparläge styrs av **ACPI**:

Global State	Name	Sleep State	Description
G0	**Working**	**S0**	Normal execution
G1	**Sleeping**	**S0ix**	Low power S0 idle
		S1	CPU caches are flushed CPU stops executing instructions
		S2	CPU is powered off
		S3	**Sleep**
		S4	**Hibernate**
G2	**Soft Off**	**S5**	Only devices that can wake the computer is powered, e.g. power-button
G3	**Mechanical Off**		Electrically off, e.g. power-cord removed

Windows Shutdown

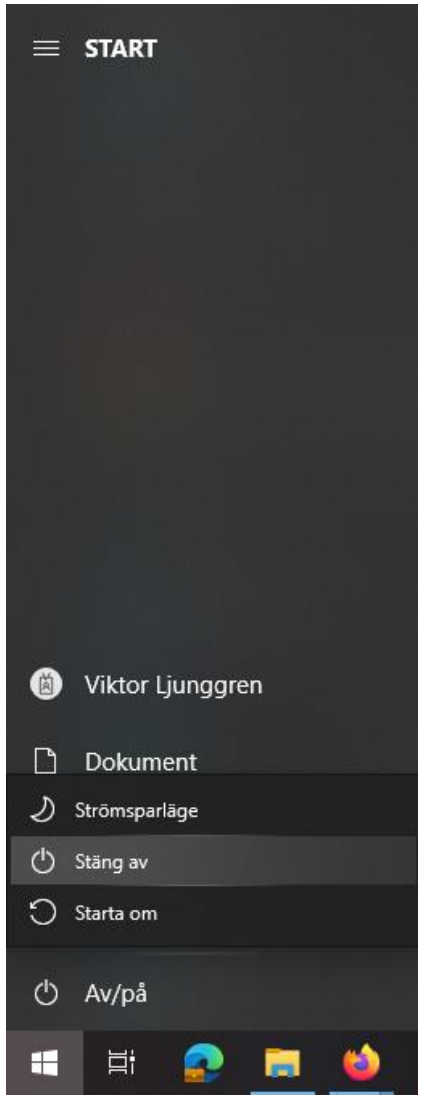


Vad händer när man stänger av?

Windows: **hibernate**

Varför?

Windows Reboot



Har du provat att starta om?

Reboot = Reboot

4. Live-demo

Caps-lock, Scroll-lock & Num-lock

CTRL-ALT-DEL

Disk timeout

Cold Reboot: Normal shutdown & startup

Soft Reboot: BIOS/UEFI reboot (CTRL-ALT-DEL), skipping POST

Graceful Shutdown: Normal shutdown e.g. with filesystem sync

Forced Shutdown: Hold power button 5-15 s. effectively a time constrained shutdown